

CRedit Author Contribution Statement

OligoVigil submission to Nucleic Acids Research, Database Issue

2026-06-11

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This statement follows the Contributor Roles Taxonomy (CRedit) as adopted by Nucleic Acids Research and Oxford University Press. The 14 standard CRedit roles are: Conceptualisation, Data Curation, Formal Analysis, Funding Acquisition, Investigation, Methodology, Project Administration, Resources, Software, Supervision, Validation, Visualisation, Writing — Original Draft, and Writing — Review & Editing.

Role assignments below follow the convention used in the team’s Octo-Agent / ESWA 2026-06-07 submission (same author group; PI-confirmed mapping).

Jie Ni — Conceptualisation; Methodology; Software; Investigation; Formal analysis; Data curation; Visualisation; Writing — original draft; Project administration.

Xinting Zhang — Investigation; Data curation; Validation.

Zhuoying Xie — Investigation; Visualisation.

Shan Lu — Resources; Writing — review & editing.

Yun Liu — Conceptualisation; Supervision; Writing — review & editing; Resources.

Adam Jatowt — Supervision; Writing — review & editing; Resources.

Notes specific to OligoVigil

- **Data Curation (Jie Ni).** Sole human curator of record (`curator_id = 'ni_jie'` in `curation_audit`). Adjudicated every one of the 1,168 firm LLM proposals plus 835 LLM-abstain cases on the 2,003 candidate pool; signed off on the 658-row human-verified release and on every override decision (20 over-accepts caught, 7 over-rejects caught, 33 abstains recovered, 92 grades adjusted).
- **Validation (Jie Ni, with Xinting Zhang).** Designed and executed the v1 forward-validation experiment (false-accept rate 0.73, 66/90, 95% CI ± 0.09 , $n = 126$); designed the 1,345-source demotion audit; confirmed the 657/658 observed-experimental coverage figure. **Inter-curator agreement (Cohen’s κ) is not claimed** because adjudication was performed by a single human curator; this is disclosed as a limitation in §Limitations(3) of the manuscript.
- **Methodology (Jie Ni).** Designed the three-stage candidate-then-LLM-then-human architecture with the candidate-vs-release firewall (the LLM never writes a curator decision); defined the toxicity ($n = 263 = 218/23/22$) and off-target ($n = 81 = 66/5/10$) benchmark splits; defined the 477 Grade A/B versus 133 not-benchmarked separation; designed the cross-split contamination audit that surfaced the 9 toxicity molecule_ids and 4 off-target source papers requiring the pair-level isolation invariant.

- **Software (Jie Ni).** Implemented the curation pipeline, the 70,283-task work queue, the v2 LLM-curator under exclusion-first + verbatim-grounding-quote constraints, the QA suites (`smoke_test.py`, `frontend_contract_check.py`, `final_delivery_check.py`), the public web portal (no-login REST API + OpenAPI + MCP + Bioschemas + W3C PROV), and the release scripts that produced `evidence_release.csv`, `benchmark_reference_splits.csv`, `curation_audit.csv`, and `v2_human_override_decisions.csv`.
- **Writing — Original Draft (Jie Ni).** Drafted the entire manuscript including the honesty-locked numbers (release 658, benchmark 344, target-gene-only 12.9%, sequence/chemistry 0%, FAR 0.73).
- **Writing — Review & Editing (all).** A post-curation copy-editing pass on the manuscript text used a large-language-model in an editorial role under a hard “numbers-and-honesty-caveats are locked” constraint (Yuan1z0825/nature-skills polishing skill, see `06_genai_disclosure.md`). All authors are responsible for the final wording.
- **Supervision (Yun Liu, Adam Jatowt).** Provided scientific oversight; reviewed all major design decisions including the candidate/release firewall and the honesty-led framing of the curation-integrity remediation.
- **Resources (Shan Lu, Yun Liu, Adam Jatowt).** Provided institutional, domain and research-computing resources used to run the curation workflows and QA suites; the computational work used the Austrian Scientific Computing (ASC) federated MUSICA cluster and the LEO5 high-performance computing facility at the University of Innsbruck (see `04_funding.md` for the full acknowledgement).

Author identities (locked, ORCID-validated)

Author	Affiliations	ORCID	Email
Jie Ni (corresp.)	SEU (1), UIBK DSC (2), NJMU (3)	0009-0003-9767-5441	njie@seu.edu.cn
Xinting Zhang	SEU (1)	0009-0005-0158-3679	xtzhang@seu.edu.cn
Zhuoying Xie	SEU (1)	0000-0003-3534-1924	zyxie@seu.edu.cn
Shan Lu	First Affil. Hosp. NJMU, Women & Children Dept (4)	0009-0004-3088-5070	lushan_sd@njmu.edu.cn
Yun Liu (corresp.)	SEU (1), First Affil. Hosp. NJMU, Information Dept (5)	0000-0002-4311-3772	liuyun@njmu.edu.cn
Adam Jatowt (corresp.)	UIBK DSC (2)	0000-0001-7235-0665	adam.jatowt@uibk.ac.at

- (1) State Key Laboratory of Digital Medical Engineering, School of Biological Science and Medical Engineering, Southeast University, Nanjing 211102, China.
- (2) Digital Science Center, University of Innsbruck, Innsbruck 6020, Austria.
- (3) Department of Medical Informatics, School of Biomedical Engineering and Informatics, Nanjing Medical University, Nanjing 211166, China.
- (4) Women and Children Department, The First Affiliated Hospital, Nanjing Medical University, Nanjing 210029, China.
- (5) Department of Information, The First Affiliated Hospital, Nanjing Medical University, Nanjing 210029, China.

All six ORCID iDs have been checksum-validated (ISO 7064 MOD 11-2). For NAR ScholarOne submission, each co-author should link the matching ORCID iD to their Editorial Manager account when invited.